

DTMF-Based FOOD DISPENSER For AQUARIUM

TEJ VIJAYKUMAR PATEL

Presented here is an Arduino-based automatic fish feeding system for an aquarium. Feeding the fish is a daily task, which can become a problem if you are out of station for more than a day. This dispenser just needs to call a cellphone number followed by another number to perform a particular task, like turning on the motor to feed the fish in the

aquarium. This project requires two cellphones, one for making the call and another for receiving it.

Circuit and working

The block diagram of the automatic food dispenser for an aquarium is shown in Fig. 1. The circuit diagram of the same is shown in Fig. 2. It consists of Arduino Uno board (Board1),

TABLE I DTMF FREQUENCIES				
F1	F2			
	1209	1336	1477	1633
697	1	2	3	A
770	4	5	6	B
852	7	8	9	C
941	*	0	#	D

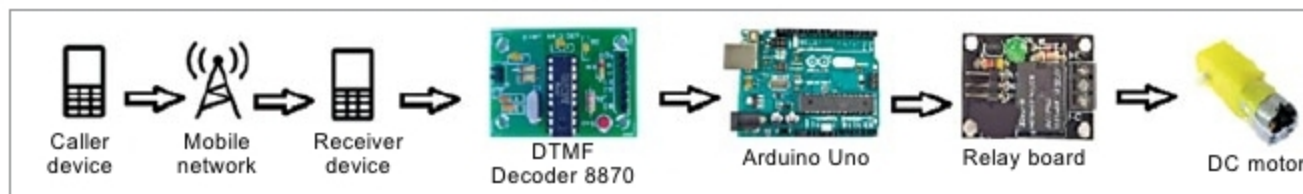


Fig. 1: Block diagram of automatic food dispensing system

DTMF decoder MT8870 (IC1), 5V voltage regulator 7805 (IC2), npn transistors BC547 (T1) and 2N2222 (T2), 5V relay (RL1), 12V DC motor and a few other components.

DTMF. Dual-tone multiple frequency (DTMF) is a signalling system for identifying the key or number dialed on a DTMF keypad. Generally, the keypad used is of 4x4 matrix type. Basically, DTMF is a sinusoidal tone, which is a combination of row and column frequencies, as shown in Table I. In this project, DTMF decoder MT8870 (IC1) is used, which is powered by 5V DC supply. It is connected with an audio jack through a single transistor (T1) amplifier stage. The jack is connected to the audio port of the cellphone.

Arduino Uno. Arduino board (Board1) is connected to IC1 and RL1, as shown in the circuit diagram. Here, pins 11 through 14 of IC1 are connected with digital I/O pins 2 through 5 of Arduino. Input signals received from IC1 are processed by Arduino.

DC motor. A geared DC motor is connected to Arduino through RL1. A single-changeover relay is used here for controlling the geared DC motor.

A special arrangement is made on the shaft of the DC

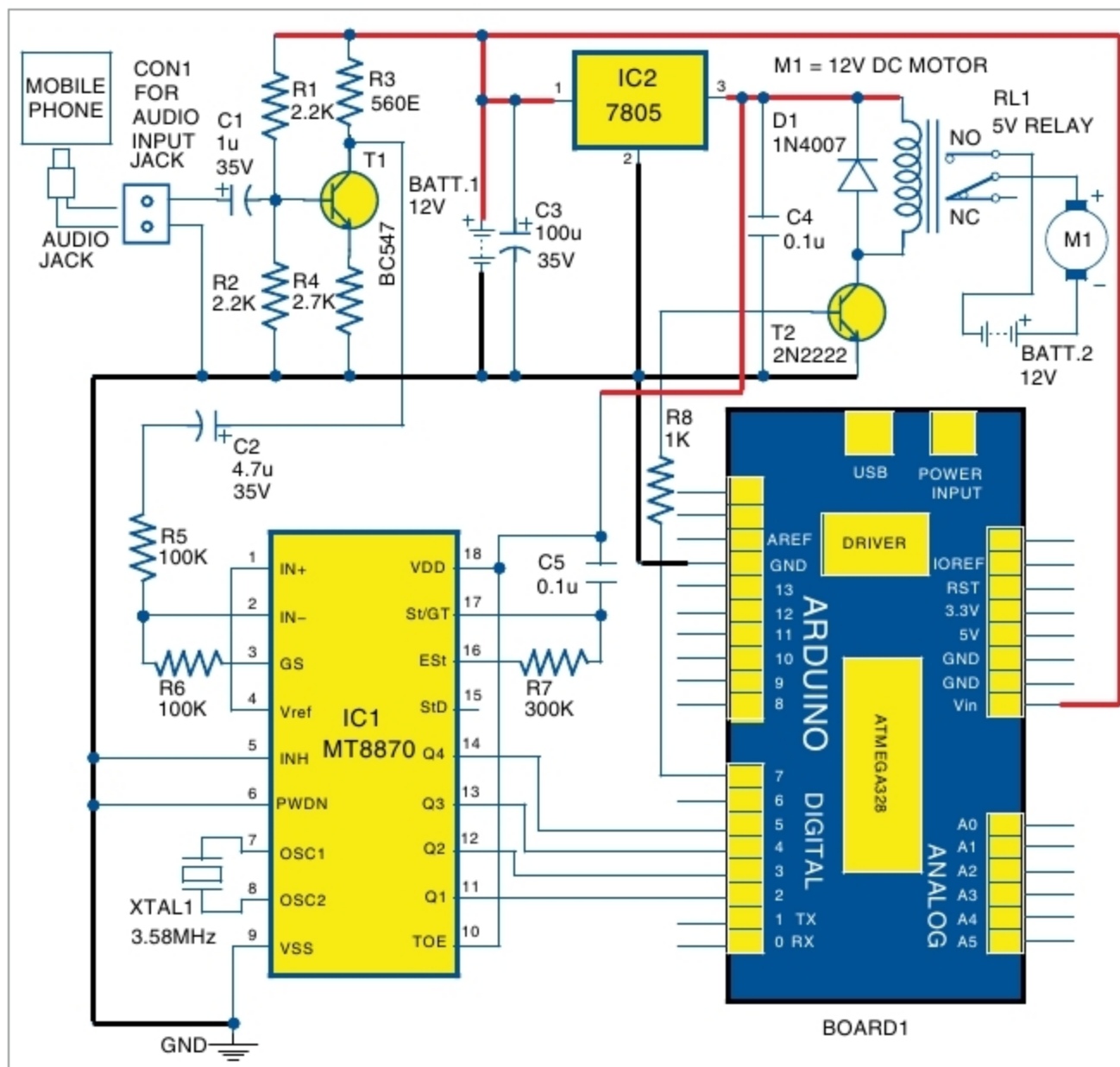


Fig. 2: Circuit diagram of automatic food dispenser for aquarium